

SOFTWARE ENGINEERING LAB-1

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**PROJECT TITLE : Personal Subscription &
Recurring Expense Auditor**

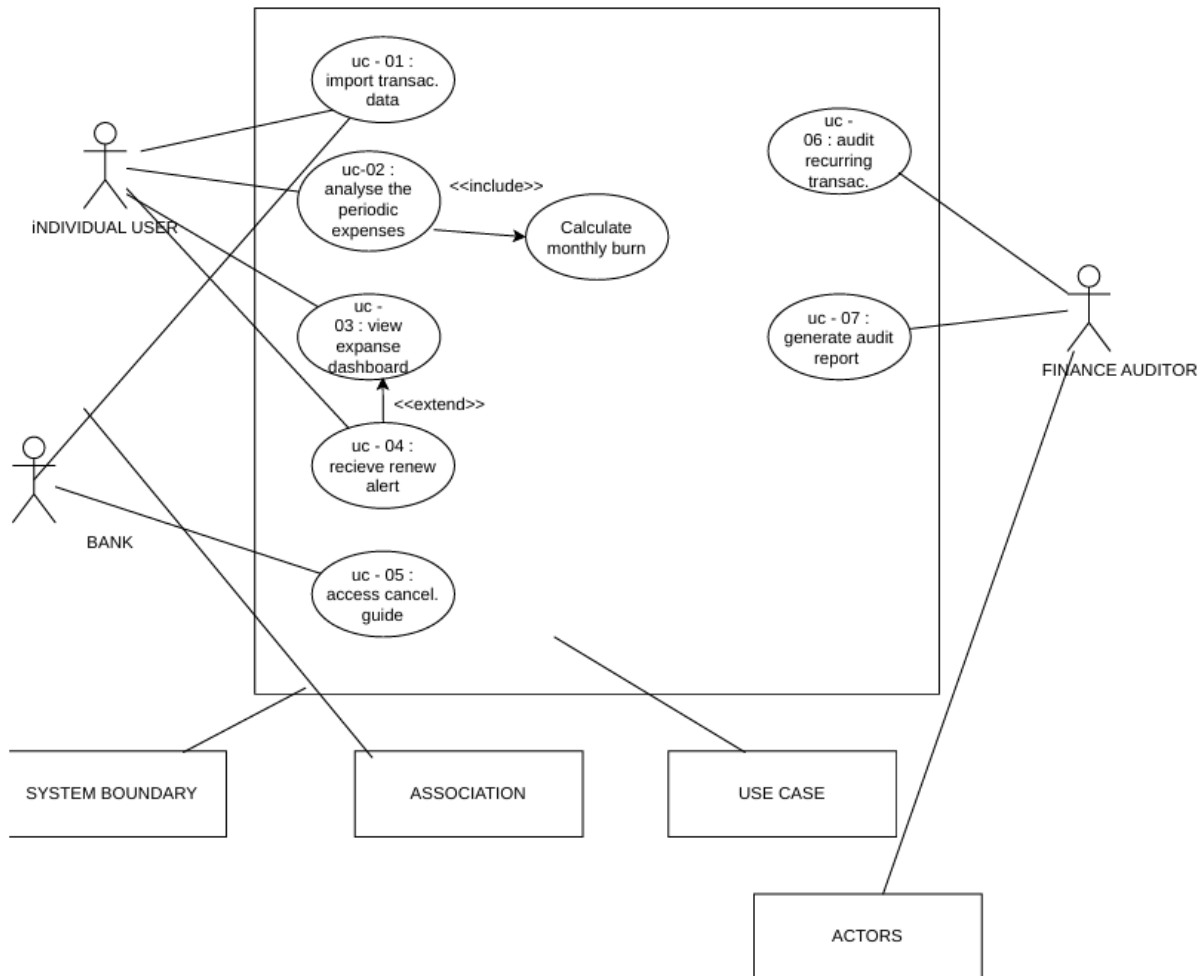
Personal Subscription & Recurring Expense Auditor

Complete Requirements Table:

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACC CRITERIA	RATIONALE	COMMENTS
FR - 001	Functional	The system shall analyze transaction logs to identify recurring billing patterns and project total monthly subscription expenditure.	High	Recurring service identified and renewal date logged	Core functionality for identifying recurring expenses	Specify the recurring-pattern detection criteria if implementation details are required.
FR - 002	Functional	The system shall allow the Individual User to import financial transaction data in CSV or OFX format	High	A valid CSV/OFX file is successfully imported and its transactions are displayed.	Transaction data is required for expense analysis.	Specify maximum supported file size if needed
FR - 003	Functional	System should alert the individual before the subscription reaches its renewal date	High	An alert is generated before the configured renewal date	Helps users avoid unwanted recurring charges	Specify how many days before renewal the alert should be generated
FR - 004	Functional	The system shall provide the Individual User with a	Medium	Selecting a subscription displays its	Supports the application's one-click	Clarify whether cancellation is performed directly or only through provided

		cancellation guide for an identified recurring subscription		applicable cancellation instructions	cancellation -guide objective.	instructions
FR - 005	Functional	The system shall allow the Finance Auditor to review recurring transactions and generate an expense audit report	High	The auditor can view identified recurring transactions and generate a report containing subscription and expenditure information	Enables financial auditing and verification	Specify the required fields in the audit report
NFR - 001	Nonfunc.	The system shall securely parse financial transaction imports without storing raw banking credentials in cleartext	High	Security testing confirms that imported credentials are not stored in cleartext	Protects sensitive financial information	Security acceptance criterion should be verified through security testing
NFR - 002	Nonfunc.	The system shall process a transaction file containing up to 10,000 transactions and display the analysis results within 5 seconds under normal operating conditions	High	A 10,000-transaction test file produces analysis results within 5 seconds	Ensures acceptable performance for realistic transaction datasets	Test using a fixed 10,000-transaction dataset to ensure repeatability

UML Use-Case Diagram:



Use-Case Flow Specification:

Use Case: Analyze Recurring Expenses

Primary Actor: Individual User

Supporting Actor: Bank / Transaction Data Source

Main Success Scenario:

1. Individual User opens the Personal Subscription & Recurring Expense Auditor.
2. User selects the "Import Transaction Data" option.
3. System displays the import options (CSV/OFX) and the user uploads a valid transaction file.
4. System validates the file format and parses the transaction data.
5. System analyzes the transaction history to identify recurring billing patterns.
6. System identifies recurring subscriptions and determines their renewal dates.
7. System calculates the projected monthly subscription expenditure.
8. System displays the list of recurring subscriptions, renewal dates, and monthly expenditure to the user.
9. User reviews the recurring subscription information.
10. Use case ends successfully.

Alternate Flow

A1. Invalid Transaction File

- A1.1. At Step 3, the system detects that the uploaded file is invalid, corrupted, or unsupported.
- A1.2. System displays an error message indicating that the file cannot be processed.
- A1.3. User is prompted to upload a valid CSV/OFX file.

- **A1.4. Use case returns to Step 3.**

A2. No Recurring Subscription Found

- **A2.1. At Step 5, the system does not find any recurring billing patterns.**
- **A2.2. System displays a message stating that no recurring subscriptions were found.**
- **A2.3. System displays the total transaction expenditure if available.**
- **A2.4. Use case ends.**

A3. Transaction Data Cannot Be Processed

- **A3.1. At Step 4, the system is unable to parse the imported transaction data.**
- **A3.2. System displays a data processing error message.**
- **A3.3. User is prompted to upload the transaction file again.**
- **A3.4. If the problem persists, the use case is cancelled.**
- **A3.5. Use case ends.**